

Dr. Doc:

Respiratory System

▼ The respiratory system is a group of organs that conduct air from the external environment to the lungs and allow gas exchange (oxygen enters the blood and carbon dioxide leaves the body).

Two Ways to Classify the Respiratory System

◆ Structural Classification (Based on Anatomy)

✓ This classification depends on the location of the organs.

◆ Upper Respiratory Tract (URT) 📌

Includes:

📌 Nose

🏠 Nasal cavity

👤 Pharynx

Note: Many anatomy texts (including Gray's and Moore) also include the 🗣️ larynx in the upper respiratory tract, although some clinical sources classify it with the lower tract. Always follow the classification used by your instructor or textbook.

✂️ Main Functions

✂️ Filters dust

💧 Humidifies air

🔥 Warms inspired air

📌 Smell (olfaction)

🛡️ First line of defense against pathogens

◆ Lower Respiratory Tract (LRT) 🫁

Includes:

 Larynx (in many clinical classifications)

 Trachea

 Bronchi

 Bronchioles

 Alveoli

 Lungs

 Main Functions

✓ Conduct air

✓ Gas exchange

✓ Voice production (larynx)

 Functional Classification

✓ Instead of looking at where the organs are, this classification looks at what they do.

There are two portions.

A.  Conducting Portion ("Air Highway")

✓ Imagine a road carrying cars .

✓ No gas exchange occurs here.

✓ Its only job is to transport and prepare the air.

 Structures

 Nose  Primary bronchi

 Nasal cavity  Secondary bronchi

-  Pharynx
-  Tertiary bronchi
-  Larynx
-  Bronchioles
-  Trachea

✗ No gas exchange occurs here.

B. Respiratory Portion ("Gas Exchange Zone")

✓ This is the business end of the respiratory system.

✓ Here, oxygen enters the blood and carbon dioxide leaves the blood .

Structures

 Respiratory bronchioles

 Alveolar ducts

 Alveolar sacs

 Alveoli

Notices

 Upper respiratory tract is all conducting

 Lower respiratory tract has both conducting and respiratory portions

 Structural classification

▼ Upper respiratory tract

▼ Lower respiratory tract

 Functional classification

▼ Conducting portion

▼ Respiratory portion

🔑 No gas exchange occurs in the conducting portion.

🔑 Gas exchange occurs only in the respiratory portion.

🔑 The alveoli 🍇 are the principal site of oxygen and carbon dioxide exchange.

🔑 The Nose (External Nose & Nasal Cavity)

🔑 The Nose Has Two Main Parts

1 External Nose 🔑

The external nose is the visible part of the respiratory system on the face.

It is supported by two frameworks:

🌟 Bony Framework (Upper One-Third)

▼ Gives rigidity and Protects the upper nose

🌟 Cartilaginous Framework (Lower Two-Thirds)

▼ Keeps nostrils open, Provides flexibility and Prevents collapse during breathing

🏠 Internal Nose (Nasal Cavity)

The nasal cavity begins at the nostrils and ends at the choanae (posterior nasal apertures), where it opens into the nasopharynx.

🎯 Important Structures of the Nasal Cavity

1 Nasal Septum

▼ Think of the septum as the wall dividing two rooms.



| Left Room |  | Right Room |

The nasal septum divides the nasal cavity into right and left halves.

2 Vestibule

✓The vestibule is the entrance hall of the nose.

✓It lies just inside the nostrils.

3 Nasal Conchae 3

These curved bony projections increase the surface area of the nasal cavity.

Three Conchae

- ◆ Superior
- ◆ Middle
- ◆ Inferior

Functions

 Create turbulent airflow

 Warm air

 Humidify air

 Filter dust

NB:- The conchae do not block airflow; they slow it and create turbulence, allowing better conditioning of inspired air.

4 Meatuses 4

A meatus is the passage beneath each concha.

Think of the conchae as bridges 🌉 and the meatuses as the roads underneath 🚗.

🏠 Superior Concha

🌉 Superior Meatus

🏠 Middle Concha

🌉 Middle Meatus

🏠 Inferior Concha

🌉 Inferior Meatus

Function

These passages allow:

🌬️ Airflow

💧 Drainage of paranasal sinuses

👉 Drainage of the nasolacrimal duct (into the inferior meatus)

🎯 Summary

🏠 The external nose consists of a bony upper one-third and a cartilaginous lower two-thirds.

🏠 The nasal septum divides the nasal cavity into right and left halves.

🏠 The vestibule is lined by skin and contains vibrissae (coarse hairs).

🏠 There are three conchae: superior, middle, and inferior.

👉 The meatuses lie below the conchae and serve as important drainage pathways for the paranasal sinuses.

Paranasal Sinuses

Paranasal sinuses are:

- ◆ Air-filled cavities (paired spaces)
- ◆ Located within four skull bones
- ◆ Lined by respiratory mucosa
- ◆ Connected to the nasal cavity through small openings (ducts or ostia)

The Four Paranasal Sinuses

1 Frontal Sinus

- 📍 Located in the frontal bone.

The sinus sits just behind your forehead.

Function

- 📍 Makes forehead lighter
- 🎤 Helps voice resonance
- 🔥 Warms inhaled air

2 Ethmoidal Sinuses

Located inside the ethmoid bone.

Unlike the others, they consist of many small air cells, like a honeycomb 🍯.

3 Sphenoidal Sinus

Located inside the sphenoid bone.

📌 Deep in the skull

🧠 Think:

🎯 Hidden cave deep behind the nose.

🚨 Clinical Importance

Very close to:

🧠 Pituitary gland

👁️ Optic nerve

🩸 Internal carotid artery

4 Maxillary Sinus

The largest paranasal sinus.

Located in the maxilla (cheek bone).

Clinical Correlation

🎯 Sinusitis

If an ostium becomes blocked 🚫:

Mucus cannot drain.



Infection develops.



😓 Facial pain

😓 Fever

 Nasal congestion

 Maxillary Sinusitis: Most common type.

Symptoms:

 Cheek pain

 Upper tooth pain

Reason:

The opening of the maxillary sinus is high on its medial wall, so drainage is less efficient.

Functions of the Paranasal Sinuses

1. Reduce Skull Weight
2. Voice Resonance
3. Warm Inspired Air
4. Humidify Inspired Air
5. Produce Mucus
6. Absorb of shock during trauma

NB:- All paranasal sinuses communicate with the nasal cavity through small openings called ostia.

 Summary

 There are four pairs of paranasal sinuses.

 They are located in the frontal, ethmoid, sphenoid, and maxillary bones.

 They are air-filled cavities lined by respiratory mucosa.

 All communicate with the nasal cavity through ostia (ducts).

🔑 The maxillary sinus is the largest paranasal sinus.

🔑 The ethmoidal sinus is composed of multiple air cells, unlike the other sinuses.

🔑 Major functions include:

- ✓ Reducing skull weight
- ✓ Adding resonance to the voice
- ✓ Warming inspired air
- ✓ Humidifying inspired air
- ✓ Absorb of shock during trauma
- ✓ Producing mucus and contributing to airway defense

Pharynx

- ▼ commonly referred to as the throat
- ▼ is a funnel-shaped passageway that serves as a common component for both the respiratory and digestive systems.
- ▼ It originates posterior to the nasal and oral cavities and extends inferiorly to the level of the cricoid cartilage, where it bifurcates into the larynx and esophagus.
- ▼ Its walls are composed of skeletal muscles used for swallowing and are lined with a mucous membrane.

▲ Shape

- ✓ Wider superiorly
- ✓ Narrower inferiorly

The pharynx is divided into three distinct anatomical regions:

1. Nasopharynx

- ▼ This is the superiormost region of the pharynx
- ▼ located posterior to the nasal cavity and superior to the soft palate.

Function:

- ◆ Normally, only air passes through this region.
- ◆ It also helps equalize air pressure between the pharynx and the middle ear.

Key Openings: It contains five openings:

- ✓ two internal nares (choanae),
- ✓ two openings for the auditory (Eustachian) tubes, and
- ✓ one opening into the oropharynx.

Associated Structures: The posterior wall contains the pharyngeal tonsil (also known as adenoids). The soft palate elevates during swallowing to block material from entering this region.

2. Oropharynx

▼ The middle region of the pharynx lies immediately posterior to the oral cavity, extending from the soft palate to the level of the hyoid bone.

Function:

It serves as a common passageway for

- ✓ air,
- ✓ food, and
- ✓ drink.

Key Openings: It has a single opening that connects to the oral cavity.

Associated Structures: It houses two pairs of tonsils that provide a "first line of defense" against ingested or inhaled materials:

- ✓ the palatine tonsils (on the lateral walls) and
- ✓ the lingual tonsils (at the base of the tongue).

3. Laryngopharynx (Hypopharynx)

▼ This is the inferior, narrowed region that begins at the level of the hyoid bone.

▼ Function:

✓ Like the oropharynx, it is a shared respiratory and digestive pathway.

Anatomical Boundaries:

✓ At its inferior end, it opens posteriorly into the esophagus

✓ Anteriorly into the larynx.

✓ It terminates at the superior border of the esophagus, which corresponds to the inferior border of the cricoid cartilage.

General Functions of the Pharynx

✓ Provides a route for both air and food.

✓ Acts as a resonating chamber for speech sounds.

✓ Houses various tonsils that support the body's immune defense.

✓ The skeletal muscles in its walls contract to force swallowed food into the esophagus.

Summary

 The pharynx is common to both the respiratory and digestive systems.

 It is funnel-shaped: wide superiorly and narrow inferiorly.

 The wall contains skeletal muscle, important for swallowing.

 Nasopharynx: only air normally passes through.

 The soft palate elevates during swallowing to close off the nasopharynx and prevent food from entering the nasal cavity.

 The auditory (Eustachian/pharyngotympanic) tubes connect the nasopharynx to the middle ear and help equalize pressure.

 The pharyngeal tonsil (adenoids) is a single lymphoid organ on the posterior wall of the nasopharynx.

 Oropharynx: common pathway for both air and food.

 The palatine tonsils lie between the palatoglossal and palatopharyngeal arches.

 The lingual tonsils are located at the base of the tongue.

 The laryngopharynx extends from the hyoid bone to the esophagus and is continuous anteriorly with the larynx.

LARYNX — Voice Box

The larynx is a short, somewhat cylindrical airway and phonatory organ that connects the laryngopharynx → trachea.

Location

Midline of the neck

Anterior to C4–C6 vertebrae

Between the laryngopharynx and trachea

Main Functions

Air passage

Conducts air into the lower respiratory tract

Protection

Prevents food/liquid from entering the airway

Phonation

Produces sound/voice

Cartilages of the Larynx

The larynx is supported by 9 cartilages:

🎯 3 Unpaired + 3 Paired = 9

Unpaired cartilages:

- ✓ Thyroid
- ✓ Cricoid
- ✓ Epiglottis

Paired cartilages:

- ▲ Arytenoid ×2
- ◆ Corniculate ×2
- ◆ Cuneiform ×2

▼ Thyroid Cartilage 🦴

- ✓ Largest laryngeal cartilage
- ✓ Forms the laryngeal prominence = Adam's apple
- ✓ Its superior border is approximately at C4
- ✓ Protects the internal structures of the larynx

🎯 Important

The two laminae meet anteriorly to form the laryngeal prominence.

▼ Cricoid Cartilage 🦴

- ✓ Located inferior to thyroid cartilage
- ✓ Located superior to the trachea
- ✓ Complete ring of cartilage

🎯 Exam point:

The cricoid cartilage is the only complete cartilaginous ring in the respiratory tract.

▼ Epiglottis

- ✓ Leaf-shaped elastic cartilage
- ✓ Located at the entrance of the larynx
- ✓ Helps protect the airway during swallowing

 During swallowing:

Food → pharynx → epiglottic mechanism → airway protected

⚠ Important correction: The epiglottis does not simply act like a lid that actively folds down over the larynx. Laryngeal closure involves coordinated movements of the laryngeal structures, including backward movement of the epiglottis.

▼ Arytenoid Cartilage

- ✓ Paired
- ✓ Pyramidal in shape
- ✓ Articulate with the cricoid cartilage
- ✓ Very important for movement of the vocal folds
- ✓ Each arytenoid has important processes:
 - ◆ Vocal process → attachment of vocal ligament
 - ◆ Muscular process → attachment of intrinsic laryngeal muscles

 Think:

Arytenoid = controls vocal fold movement 

▼ Corniculate Cartilages ◆

- ✓ Paired
- ✓ Small, conical-shaped cartilages
- ✓ Located on the apices of arytenoid cartilages
- ✓ Support the posterior part of the laryngeal inlet

▼ Cuneiform Cartilages ◆

Paired

Small, rod-shaped cartilages

Located within the quadrangular membrane

Support the vestibular folds and lateral laryngeal wall

👤 Vocal Folds

- ✓ The mucosa of the larynx forms two pairs of folds.

● Superior pair → Vestibular folds

- ✓ Also called: False vocal cords
- ✓ Located superiorly
- ✓ Help protect the airway
- ✓ Contribute to closure of the laryngeal inlet
- ✗ Not primarily responsible for normal voice production

● Inferior pair → Vocal folds

- ✓ Also called: True vocal cords
- ✓ Located inferiorly
- ✓ Vibrate when air passes through them
- 🎤 Produce sound

🔥 Glottis

Be careful with terminology here.

Glottis =

Vocal folds + the space between them

The space between the true vocal folds is specifically called:

👉 Rima glottidis

Distinction:

Glottis = vocal folds + opening

Rima glottidis = opening between vocal folds

Larynx — Muscles & Neurovasculature

👤 Muscles

Intrinsic muscles → control the position and tension of vocal folds 🎤

🎯 Cricothyroid → tenses/stretches vocal folds → high-pitched voice 🎵

🎯 Posterior cricoarytenoid → abducts vocal folds → opens airway (ONLY ABDUCTOR of the vocal folds)....

Extrinsic muscles → move and stabilize the larynx.

🩸 Blood Supply

Arteries: Superior & inferior laryngeal arteries ●

Veins: Superior & inferior laryngeal veins ●

🧠 Innervation

Vagus nerve (CN X) → via laryngeal nerves

✓ Superior laryngeal nerve

- ▼ External branch → motor to cricothyroid
- ▼ Internal branch → sensory above vocal folds

✓ Recurrent laryngeal nerve

- ▼ Motor to all intrinsic muscles except cricothyroid
- ▼ Sensory below vocal folds

★ Exam Point

Cricothyroid → External laryngeal nerve 🎵

All other intrinsic muscles → Recurrent laryngeal nerve ●

Posterior cricoarytenoid → ONLY abductor 🔓

🧠 Trachea

📍 Location & Extent

- ✓ Trachea = windpipe
- ✓ Part of the conducting zone
- ✓ About 10–11 cm long
- ✓ Begins at the cricoid cartilage (C6)
- ✓ Descends in front of the esophagus
- ✓ Ends at the sternal angle (T4–T5), where it divides into the main bronchi.

▼ Direction:

Larynx → Trachea → Right & Left main bronchi

▼ Structure

- ✓ 15–20 C-shaped rings of hyaline cartilage
- ✓ Rings keep the airway open (patent).
- ✓ The open posterior part faces the esophagus.
- ✓ Posterior gap is closed by a fibromuscular membrane containing the trachealis muscle
- ✓ Anular ligaments connect adjacent cartilage rings.

▼ Lining

Pseudostratified ciliated columnar epithelium + mucus-producing cells

- ✓ Traps particles
- ✓ Cilia move mucus upward ↑
- ✓ Helps clean and protect the airway

🧠 Mucociliary escalator

▼ Bifurcation

The trachea divides into:

- ➡ Right main bronchus
- ➡ Left main bronchus

★ Carina

- ✓ Internal keel-shaped ridge at the tracheal bifurcation.

✓ Highly sensitive.

✓ Stimulation can trigger the cough reflex 🤧

📌 Bifurcation: sternal angle / T4–T5

🩸 Neurovascular Supply

● Arterial Supply

Mainly from:

✓ Inferior thyroid arteries

● Venous Drainage

✓ Inferior thyroid venous plexus

→ Brachiocephalic veins

🧠 Innervation

✓ Parasympathetic: Vagus nerve (CN X), mainly via recurrent laryngeal nerves

✓ Sympathetic: Sympathetic trunks

Bronchial Tree

The bronchial tree is a branching system of air passages that conducts air from the trachea to the lungs. 🗺️

▼ Branching Order

Trachea



Primary (main) bronchi



Secondary (lobar) bronchi



Tertiary (segmental) bronchi



Bronchioles



Terminal bronchioles

➡ Terminal bronchioles = end of conducting zone

▼ Primary (Main) Bronchi

💡 The trachea divides at the sternal angle (T4–T5) into:

🫁 Right main bronchus

🫁 Left main bronchus

They contain hyaline cartilage that helps maintain airway patency.

★ Right vs Left

Feature:-	Right main bronchus	Left main bronchus
Length:-	Shorter	Longer
Diameter:-	Wider	Narrower
Direction:-	More vertical	More oblique
Aspirated foreign body:-	More common	Less common

💡 Clinical point: Foreign bodies are more likely to enter the right main bronchus because it is shorter, wider, and more vertical.

Secondary (Lobar) Bronchi

Each primary bronchus divides into lobar bronchi, with one bronchus supplying each lung lobe.

 Right lung

3 lobes → 3 lobar bronchi

- ✓Superior

- ✓Middle

- ✓Inferior

 Left lung

2 lobes → 2 lobar bronchi

- ✓Superior

- ✓Inferior

 Lobar bronchus → supplies a lobe

▼ Tertiary (Segmental) Bronchi

- ✓Arise from lobar bronchi

- ✓Supply bronchopulmonary segments

 - ◆ Right lung: usually 10

 - ◆ Left lung: usually 8–10

 Segmental bronchus → bronchopulmonary segment

Bronchioles

Segmental bronchi



Smaller bronchi



Bronchioles



Terminal bronchioles

★ Terminal bronchioles are the final branches of the conducting zone.

🫁 Respiratory Zone

✓ After terminal bronchioles:

Respiratory bronchioles



Alveolar ducts



Alveolar sacs



Alveoli 🫧

➡ Alveoli = main site of gas exchange $\leftrightarrow O_2$

Respiratory Zone

▼ The respiratory zone is the part of the lower respiratory tract where gas exchange occurs.

📌 It begins after the terminal bronchioles of the conducting zone.

Pathway

Terminal bronchiole



Respiratory bronchiole



Alveolar duct



Alveolar sac



Alveoli

Respiratory Bronchioles

✓ First part of the respiratory zone.

✓ Have small alveoli in their walls.

★ Gas exchange begins here.

🧠 Key point:

Respiratory bronchiole = first site of gas exchange.

Alveolar Ducts

✓ Arise from respiratory bronchioles.

✓ Walls are largely surrounded by alveoli.

✓ Conduct air toward alveolar sacs.

🔴 Participate in gas exchange.

Alveoli

✓ Tiny air sacs where most gas exchange occurs.

✓ Give lungs their spongy appearance

✓ Approximately 300–400 million alveoli.

✓ Diameter: about 0.25–0.5 mm

Cells of Alveoli

Type I pneumocytes

- ▼ Thin simple squamous cells → gas exchange

Type II pneumocytes

- ▼ Produce surfactant → reduces surface tension

Alveolar macrophages

- ▼ Remove particles and microorganisms

Alveolar Pores

- ▼ Small openings between adjacent alveoli.
- ▼ Allow communication of air between neighboring alveoli.

Respiratory Membrane

The respiratory membrane is the thin barrier across which O_2 and CO_2 diffuse.

Three main components:

- ✓ Type I alveolar epithelium
- ✓ Fused basement membranes
- ✓ Capillary endothelium

Clinical: Pneumonia

Pneumonia = infection/inflammation involving the lung parenchyma, commonly including alveoli.

 Bacteria / viruses / fungi



 Inflammation



 Fluid + inflammatory cells in alveoli



Effective gas-exchange area



 Impaired oxygenation

Common features:

 Cough

 Fever

 Rapid breathing

Possible shortness of breath

 Lungs

✓The lungs are paired, cone-shaped respiratory organs located in the thoracic cavity.

✓They are separated by the mediastinum and are responsible for gas exchange.

▲ General Features

◆ Apex

✓Rounded/blunt superior end.

✓Projects into the root of the neck.

✓Extends slightly above the medial clavicle.

◆ Base

✓ Broad, concave inferior surface.

✓ Rests on the diaphragm.

◆ Surfaces

Key feature

Costal

✓ Convex; contacts ribs & intercostal muscles

Mediastinal

✓ Faces mediastinum; contains hilum

Diaphragmatic

✓ Inferior surface/base; rests on diaphragm

◆ Hilum

The hilum is the area on the mediastinal surface where structures enter and leave the lung.

Structures include:

👉 Bronchi

👉 Pulmonary arteries

👉 Pulmonary veins

👉 Nerves

👉 Lymphatic vessels

➔ Collectively, these structures form the root of the lung.

Right vs Left Lung

Feature	Right lung	Left lung
Size	Larger & heavier	Smaller & lighter
Shape	Shorter & broader	Longer & narrower
Lobes	3	2
Fissures	2	1
Fissures	Horizontal + oblique	Oblique
Cardiac notch	✗ Absent	✓ Present
Lingula	✗	✓ Present

Root of the Lung

The root contains structures connecting the lung to the mediastinum.

Important arrangement

- ✓ Bronchus → posterior
- ✓ Pulmonary artery → superior
- ✓ Pulmonary veins → anterior/inferior

RALS for pulmonary artery position:

Right lung → pulmonary artery is Anterior to bronchus

Left lung → pulmonary artery is Superior to bronchus

Bronchopulmonary Segments

A bronchopulmonary segment is a relatively independent functional unit of the lung.

Each segment contains:

- ▼ Segmental (tertiary) bronchus
- ▼ Segmental pulmonary artery
 - ✓ Right lung → usually 10 segments
 - ✓ Left lung → usually 8–10 segments

🚨 Clinical importance

Because segments are relatively independent, they can be surgically removed individually and infection may remain relatively localized.

Pleura

▼ The pleura is a double-layered serous membrane surrounding each lung. It consists of mesothelium (simple squamous epithelium) and underlying connective tissue.

▼ Layers of Pleura

👤 Visceral Pleura

- ✓ Closely covers the surface of the lung
- ✓ Extends into the fissures
- ✓ Insensitive to pain
- ✓ Receives its blood supply mainly from the bronchial circulation

👤 Parietal Pleura

Lines the:

- ✓ Thoracic wall → Costal pleura
- ✓ Mediastinum → Mediastinal pleura

- ✓ Diaphragm → Diaphragmatic pleura
- ✓ Root of neck → Cervical pleura (cupula)

🎯 Parietal pleura is pain-sensitive because it has somatic sensory innervation.

Pleural Cavity

- ▼ is the potential space between the visceral and parietal pleura.
- ▼ Contains a small amount of serous fluid 💧

Function:

- ✓ Lubricates pleural surfaces
- ✓ Reduces friction during breathing
- ✓ Helps lungs move smoothly with the thoracic wall

Pleural Recesses

- ▼ These are spaces where the parietal pleura extends beyond the lung margins, providing room for lung expansion.
- ▼ Part of pleural cavity not filled with the lung
- ▼ Allow expansion of the lung during forced inspiration

◆ Costomediastinal Recess

- ✓ Between: Costal + mediastinal pleura
- ✓ More prominent on the left, because of the cardiac notch.

◆ Costodiaphragmatic Recess

- ✓ Between: Costal + diaphragmatic pleura
- ✓ Largest pleural recess

- ✓ Important site where fluid can accumulate.

Innervation

◆ Parietal Pleura

Region	Nerve
Costal pleura	Intercostal nerves
Peripheral diaphragmatic pleura	Intercostal nerves
Mediastinal pleura	Phrenic nerve
Central diaphragmatic pleura	Phrenic nerve

- ▲ Phrenic nerve irritation → referred pain to the shoulder 🦵

◆ Visceral Pleura

- ✓ Autonomic innervation associated with the pulmonary plexus
- ✓ Insensitive to pain, temperature, and touch.

🩸 Blood Supply

◆ Parietal Pleura

Supplied mainly by:

- 🩸 Intercostal arteries
- 🩸 Internal thoracic arteries
- 🩸 Musculophrenic arteries

◆ Visceral Pleura

🔴 Mainly supplied by bronchial arteries.

Clinical Conditions

Abnormalities within the pleural cavity can severely impact respiratory function:

◆Pneumothorax: The presence of free air in the pleural cavity (often from injury), which can lead to a collapsed lung, or atelectasis.

◆Pleural Effusions: Accumulation of various fluids in the cavity, including blood (hemothorax), serous fluid (hydrothorax), or pus (empyema)

Blood Supply of the Lungs

The lungs have two separate circulations:

● Pulmonary circulation → gas exchange

● Bronchial circulation → nutrition of lung tissue

● 1. Pulmonary Circulation

🔵 Deoxygenated blood → lungs

Right ventricle



Pulmonary trunk



Right & left pulmonary arteries



Lobar arteries



Segmental arteries



Pulmonary capillaries around alveoli

 At the alveoli

O₂: Alveoli → blood ↑

CO₂: Blood → alveoli ↑

➡ Blood becomes oxygenated.

♥ Oxygenated blood → heart

Pulmonary capillaries



Pulmonary veins



Left atrium

★ Usually 2 pulmonary veins from each lung:

◆ Superior pulmonary vein

◆ Inferior pulmonary vein

 Pulmonary arteries = deoxygenated

 Pulmonary veins = oxygenated

● 2. Bronchial Circulation

The bronchial arteries supply oxygenated blood to the lung tissues themselves.

🔴 Bronchial arteries supply:

- ✓ Bronchi
- ✓ Supporting connective tissue
- ✓ Structures of the root of the lung
- ✓ Visceral pleura

◆ Their blood comes from the systemic circulation, mainly through branches of the thoracic aorta.

▼ Important

- ✓ Some bronchial venous blood drains into the pulmonary veins.
- ✓ Therefore, a small amount of deoxygenated bronchial blood mixes with oxygenated pulmonary venous blood.

🫁 Bronchopulmonary Segment Blood Supply

Each bronchopulmonary segment has:

● Arteries

- ✓ Segmental pulmonary artery + Segmental bronchial artery
- ➔ Run within the segment.

● Pulmonary veins

- ✓ Usually run between adjacent segments
- ✓ Called intersegmental veins

Clinical significance

▼ Because pulmonary veins are mainly intersegmental, surgeons can use these planes to perform segmental lung resection while preserving major vessels of neighboring segments.

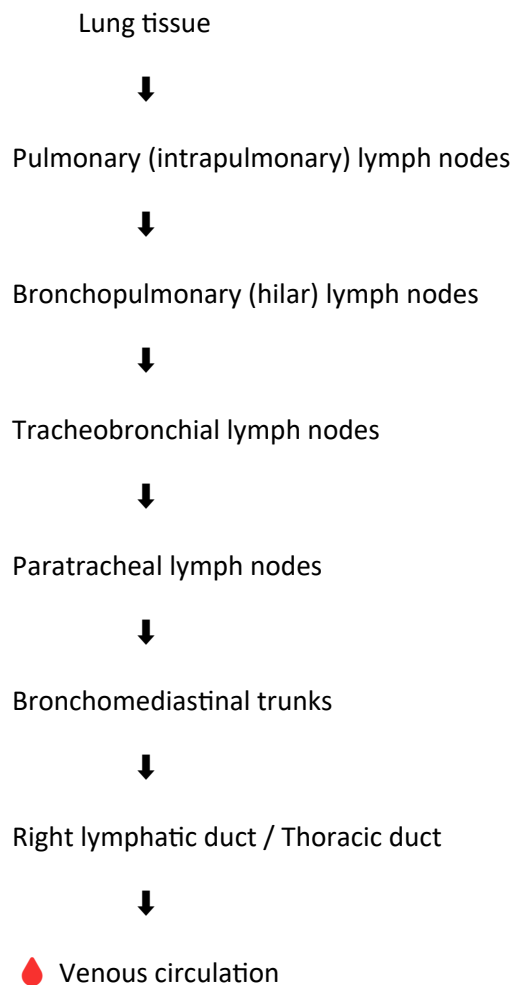
Lymphatic Drainage of the Respiratory System

▼ The lymphatic system of the lungs drains lymph, foreign particles, dust, and cellular debris from the lung tissue and airways.

● Lymph Node Groups

✓ Lymphatic vessels are found around the bronchi, within lung connective tissue, and in the pleura.

✓ Main drainage pathway



◆ Pulmonary Lymph Nodes

- ✓ Located along the bronchi within the lung.
- ✓ Receive lymph from lung tissue.
- ✓ Collect dust and carbon particles that escape the mucociliary clearance system.

♥ This explains why carbon particles can accumulate in pulmonary lymph nodes, especially with chronic exposure to air pollution or smoke.

◆ Bronchopulmonary (Hilar) Nodes ■

- ✓ Located at the hilum of the lung.
- ✓ Receive lymph from the pulmonary lymph nodes.
- ✓ Lymph then passes toward the tracheobronchial nodes.

◆ Pulmonary → Bronchopulmonary

◆ Tracheobronchial Nodes

- ✓ Located around the tracheal bifurcation (carina).
- ✓ They receive lymph from the bronchopulmonary nodes and drain toward the paratracheal nodes.

◆ Paratracheal Nodes

Located alongside the trachea.

- ✓ Receive lymph from tracheobronchial nodes
- ✓ Drain toward the bronchomediastinal trunks.

🔴 Final Drainage

Right side

Right bronchomediastinal trunk

→ usually right lymphatic duct

→ right venous angle

Left side

Left bronchomediastinal trunk

→ usually thoracic duct

→ left venous angle

⚠ The bronchomediastinal trunks can sometimes drain directly into the venous system rather than through a lymphatic duct.

Pleural Lymphatics

★ Visceral pleura

- ▼ Lymphatic vessels are closely associated with those of the lung.
- ▼ Lymphatic vessels enter/leave the lung mainly through the hilum.

★ Parietal pleura

- ▼ Has lymphatic drainage associated with the thoracic wall and diaphragm.

▼ The pulmonary ligament is a double fold of pleura that connects the lower medial part of each lung to the mediastinum. It is not a true fibrous ligament, but rather a continuous reflection where the parietal and visceral pleura meet below the root of the lung.

▼ The hilum of the lung is a wedge-shaped depression on the medial (inner) surface of each lung. It acts as the gateway or "root" where major structures like bronchi, pulmonary blood vessels, nerves, and lymphatics enter and exit the organ.

Thorax

- ✓ The thorax is the superior part of the trunk, located between the neck and abdomen.
- ✓ It contains and protects major organs of the respiratory and cardiovascular systems.

📌 Boundaries of the Thorax

Direction	Boundary
↑ Superior	Superior thoracic aperture (thoracic inlet)
↓ Inferior	Inferior thoracic aperture, closed by the diaphragm
● Anterior	Sternum
● Posterior	12 thoracic vertebrae + intervertebral discs
↔ Lateral	12 pairs of ribs + intercostal muscles

● The diaphragm separates: Thoracic cavity and Abdominal cavity 🧑

Thoracic Apertures

1 Superior Thoracic Aperture

- ✓ Also called thoracic inlet
- ✓ Connects the thorax with the neck.
- ✓ Structures such as the trachea, esophagus, major vessels, and nerves pass through it.

2 Inferior Thoracic Aperture

- ✓ Larger than the superior aperture.

- ✓ Normally closed by the diaphragm.
- ✓ Allows passage of structures between thorax and abdomen through openings in the diaphragm.

Thoracic Cavity

The thoracic cavity is divided into 3 major compartments:

1. Right Pleural Cavity

➡ Contains the right lung

2. Left Pleural Cavity

➡ Contains the left lung

3. Mediastinum

➡ Central compartment between the two pleural cavities.

Functions of the Thorax

The thorax has 4 major functions:

Breathing → Protection → Support → Conduit

▼ Breathing 🗨️

The thorax provides the mechanical system needed for ventilation.

💪 Diaphragm

- ✓ Principal muscle of respiration
- ✓ Dome-shaped musculotendinous muscle.
- ✓ Contraction → moves inferiorly ↓
- ✓ Thoracic volume ↑ → air enters the lungs (inspiration).

Thoracic Wall & Ribs

- ✓ Movement of the ribs changes the anteroposterior and transverse dimensions of the thorax.
- ✓ Costovertebral joints allow rib movement during respiration.

Thoracic volume ↑ → inspiration

Thoracic volume ↓ → expiration

◆ Protection

The thoracic cage forms an osteocartilaginous protective framework.

Protects:

- ✓ Heart
- ✓ Lungs
- ✓ Great vessels

Also protects some abdominal organs:

- ▼ Liver → mainly beneath the right dome of diaphragm
- ▼ Stomach & spleen → beneath the left dome
- ▼ Superior poles of kidneys → partly protected by lower ribs

- ▼ Spinal cord

The thoracic vertebrae and their long, inferiorly directed spinous processes contribute to posterior protection.

◆ Support

The thoracic cage provides attachment and support for:

Upper limbs

- ▼ Provides attachment sites for muscles connecting the thorax to the upper limb.
- ▼ Helps support the weight of the upper limbs.

Muscles

Ribs, sternum, and thoracic vertebrae provide attachment for muscles of:

- ▼ Back
- ▼ Chest
- ▼ Neck
- ▼ Shoulders
- ▼ Abdomen

★ Some of these muscles also act as accessory muscles of respiration.

◆ Conduit

The mediastinum acts as a passageway for structures traveling between the neck, thorax, and abdomen.

Important structures passing through the thorax:

- | | |
|----------------------|------------------------------|
| ▼ Esophagus | ▼ Vagus nerves |
| ▼ Thoracic duct | ▼ Phrenic nerves |
| ▼ Trachea | ▼ Thoracic aorta |
| ▼ Superior vena cava | ▼ Superior Thoracic Aperture |

The thoracic inlet allows structures to pass between:

Neck ↔ Thorax

Thoracic Cage

▼ The thoracic cage is an osteocartilaginous framework that provides protection, support, and mechanical assistance in respiration.

Components

The thoracic cage consists of:

- ▼ 12 thoracic vertebrae
- ▼ 12 pairs of ribs
- ▼ Costal cartilages
- ▼ Sternum

Ribs

Ribs are curved, flat bones forming most of the thoracic cage.

Classification by Sternal Attachment

Type	Ribs	Attachment
 True ribs	1–7	Directly attach to sternum via their own costal cartilage
 False ribs	8–10	Attach indirectly through cartilage of the rib above
 Floating ribs	11–12	No anterior attachment

🌟 Typical Ribs — 3rd–9th

A typical rib has:

Head → Neck → Tubercle → Shaft

1 Head

✓Has 2 articular facets separated by a crest.

✓Superior facet → vertebra above

✓Inferior facet → corresponding vertebra

2 Neck

✓Short, flattened portion.

✓Connects head → tubercle.

3 Tubercle

Has:

✓Articular part → transverse process of corresponding vertebra

✓Non-articular part → ligament attachment

4 Shaft

✓Long, thin, curved.

✓Has a costal angle.

✓Internal surface contains the costal groove.

★ Costal Groove

The costal groove protects the:

🔴 Vein

🔴 Artery

🧠 Nerve

➡ VAN runs along the inferior border of the rib.

Atypical Ribs

▼ 1st Rib

The 1st rib is:

▼ Shortest

▼ Broadest

▼ Most curved

▼ Almost horizontal

Important features:

▼ Single facet on head → T1

▼ Superior surface has 2 grooves:

✓ Subclavian vein

✓ Subclavian artery

▼ Grooves are separated by the scalene tubercle 💪

✖ 1st rib = Subclavian vessels + scalene tubercle

▼ 2nd Rib

✓ Longer and thinner than 1st rib.

✓ Has a distinctive tuberosity for serratus anterior.

✖ 2nd rib → Serratus anterior 💪

▼ 10th Rib

▼ Usually has one articular facet on its head.

▼ Articulates with T10.

11 & 12 Ribs

Both are floating ribs 🌊.

They:

✓ Have single facet on head.

✓ Lack neck

✓ Lack tubercle

✓ Have no anterior attachment

◆ Sternum

▼ The sternum is a flat, elongated bone in the anterior midline.

It has 3 parts:

Manubrium → Body → Xiphoid process

🏹 Manubrium

▼ Superior and broadest part.

Contains:

🎯 Jugular (suprasternal) notch

 Clavicular notches → articulate with clavicles

 Approximately T3–T4.

 Body

✓ Longest part of sternum.

✓ Formed from fused sternebrae.

✓ Approximately T5–T9.

 Xiphoid Process

✓ Smallest and most variable part.

✓ Often cartilaginous in young people.

✓ Usually ossifies progressively during adulthood.

★ Sternal Angle — Angle of Louis

The sternal angle is the junction between:

 Manubrium + body

 T4–T5 level

Marks:

 2nd costal cartilage

 Tracheal bifurcation

 Beginning/end of important thoracic structures at the transverse thoracic plane

Thoracic Vertebrae

There are 12 thoracic vertebrae (T1–T12).

★ Characteristic features

♥ Vertebral body

✓ Heart-shaped

✓ Larger than cervical vertebral bodies.

▣ Spinous process

✓ Long

✓ Slants inferiorly

➡ Helps protect the spinal cord.

Costal facets

✓ Thoracic vertebrae articulate with ribs.

On vertebral body:

Typical thoracic vertebrae have:

▼ Superior demifacet

▼ Inferior demifacet

➡ Receive the head of adjacent ribs.

Special vertebrae:

▼ T1 → has a complete facet for rib 1 + demifacet for rib 2

▼ T10 → usually one complete facet for rib 10

▼ T11–T12 → one complete facet for their corresponding ribs

On transverse processes:

- ▼ Typical T1–T10 → facet for rib tubercle
- ▼ T11 & T12 → NO transverse costal facet ❌

Thoracic Vertebral Foramen

- ✓ Usually small and circular
- ✓ Smaller than cervical vertebral foramina.

Thoracic Apertures

There are 2 major openings:

1 Superior Thoracic Aperture — Thoracic Inlet

- ▼ Connects the thorax ↔ neck.

Boundaries:

← BACK Posterior: T1 vertebra

↔ Lateral: 1st ribs + costal cartilages

● Anterior: Superior border of manubrium

Function:

Allows passage of structures between the neck and thorax, including:

👤 Trachea

👤 Esophagus

👤 Major vessels

👤 Nerves

2 Inferior Thoracic Aperture — Thoracic Outlet

- ▼ Much larger than the superior aperture.

Boundaries:



Posterior: T12 vertebra

↔ Posterolateral: 11th & 12th ribs

↔ Anterolateral: Costal margins (ribs 7–10)



Anterior: Xiphisternal joint

Notices

- ▼ The inferior thoracic aperture is closed by the diaphragm.
- ▼ The diaphragm separates: Thoracic cavity and Abdominal cavity



Thoracic Wall

◆ The thoracic wall is a musculoskeletal framework that provides protection, support, and respiratory movements.



1. Joints of the Thoracic Wall

These joints allow the thoracic cage to expand and contract during breathing. 🗨️

1 Intervertebral Joints

- 📌 Between the bodies of adjacent thoracic vertebrae.
- ➡ Allow limited movement of the vertebral column.

2 Costovertebral Joints

- ▼ These connect the ribs to the vertebral column.

▼ A typical rib has two main articulations:

A. Joint of Head of Rib

Rib head $\leftrightarrow$ vertebral bodies

B. Costotransverse Joint

Rib tubercle $\leftrightarrow$ transverse process

➡ Together, these joints allow the ribs to move during respiration.

3 Sternocostal Joints

Connect costal cartilages $\rightarrow$ sternum.

▼ 1st sternocostal joint $\rightarrow$ primarily cartilaginous

▼ 2nd–7th $\rightarrow$ synovial joints

➡ Permit movement of the ribs during breathing.

4 Manubriosternal Joint

▼ Located between: Manubrium $\leftrightarrow$ Body of sternum

▼ Forms the palpable sternal angle (Angle of Louis).

▼ Approximately T4–T5

▼ Marks the level of the 2nd costal cartilage.

5 Xiphisternal Joint

▼ Located between: Body of sternum $\leftrightarrow$ Xiphoid process

▼ Important anterior thoracic landmark.

6 Other Thoracic Joints

 Costochondral

Rib ↔ Costal cartilage

 Interchondral

Adjacent costal cartilages

 Sternoclavicular

Clavicle ↔ Manubrium

2. Muscles of the Thoracic Wall

The intrinsic muscles of the thoracic wall are mainly arranged in three intercostal layers.

◆ External Intercostals

▼ Superficial layer

Action:

↑ Elevate ribs

➡ Important during inspiration

 External = Elevate

◆ Internal Intercostals

▼ Middle layer

Action:

↓ Depress ribs

➡ Particularly important during forced expiration.

🧠 Internal = Inferior movement

◆ Innermost Intercostals

▼ Deepest intercostal layer.

Action:

↓ Help depress ribs

➡ Function similarly to internal intercostals.

★ Important anatomical relationship

The intercostal neurovascular bundle lies between:

Internal intercostal + innermost intercostal muscles

💪 Other Intrinsic Muscles

[4] Subcostales

Located mainly in the lower thoracic wall.

Help depress the ribs.

[5] Transversus Thoracis

📌 Runs from posterior sternum → costal cartilages 2–6.

Action:

↓ Weakly depresses the costal cartilages.

[6] Levatores Costarum

Elevate the ribs.

Assist inspiration.

Accessory Respiratory Muscles

Several muscles outside the intrinsic thoracic wall can assist respiration, especially during forced breathing.

● Pectoral muscles

Pectoralis major

Pectoralis minor

● Serratus anterior

➡ Can assist elevation of ribs when the upper limb is fixed.

● Abdominal muscles

➡ Important in forced expiration.

● Back muscles

Latissimus dorsi

Rhomboids

➡ Can assist respiratory movements under certain conditions.

Diaphragm — Principal Muscle

▼ The diaphragm is the principal muscle of respiration.

▼ It is a dome-shaped musculotendinous partition between:

◆ Thorax ↔ Abdomen

During inspiration:

Diaphragm contracts



Moves inferiorly



Thoracic volume ↑



Air enters lungs 🌬️

During quiet expiration:

Diaphragm relaxes



Moves superiorly



Thoracic volume ↓



Air leaves lungs

The thoracic wall is supplied segmentally by the intercostal nerves, arteries, and veins that run in the intercostal spaces.

1. Innervation

There are 12 pairs of thoracic spinal nerves.

Intercostal nerves

▼ T1–T11 anterior rami → intercostal nerves

▼ T12 anterior ramus → subcostal nerve

They run mainly between the internal and innermost intercostal muscles.

🔥 Main branches

Branch	Supplies
Muscular branches	Intercostal muscles, subcostals, transversus thoracis
Lateral cutaneous	Skin of lateral thorax & abdomen
Anterior cutaneous	Skin of anterior thorax & abdomen
Collateral branches	Intercostal muscles & parietal pleura
Posterior rami	Back muscles, joints, bones & skin

Important

Intercostal nerves = anterior rami of T1–T11

T12 = subcostal nerve 🎯

2. Arterial Supply

Each typical intercostal space receives:

1 posterior intercostal artery

2 anterior intercostal arteries

➡ Therefore, 3 arterial sources per typical intercostal space.

● Posterior Intercostal Arteries

1st–2nd spaces

➡ From supreme intercostal arteries, branches of the costocervical trunk from the subclavian artery.

3rd–11th spaces

➡ Direct branches of the thoracic aorta.

1–2 → Subclavian system

3–11 → Aorta

● Anterior Intercostal Arteries

1st–6th spaces

➡ Internal thoracic artery

7th–9th spaces

➡ Musculophrenic artery

10th–11th spaces

✗ No anterior intercostal arteries

➡ Supplied by posterior intercostal arteries.

3. Venous Drainage

Intercostal veins accompany the arteries and nerves.

★ Position in the main neurovascular bundle:

✓ Vein

✓ Artery

✓ Nerve

The bundle lies in the costal groove along the inferior border of the rib.

● Anterior Intercostal Veins

Drain mainly into:

➡ Internal thoracic veins

➡ Musculophrenic veins

● Posterior Intercostal Veins

Drain mainly into the azygos system.

Right side → ■ Azygos vein

Most right posterior intercostal veins

➡ Azygos vein

➡ SVC

Left side → ■ Hemiazygos system

Most left posterior intercostal veins

➡ Hemiazygos / accessory hemiazygos

➡ Azygos

➡ SVC

Special: 1st Posterior Intercostal Veins

The 1st posterior intercostal veins usually drain directly into:

➡ Brachiocephalic veins

Diaphragm

🌀 The diaphragm is a dome-shaped musculetendinous partition separating the thoracic cavity 🫁 from the abdominal cavity 🧑 .

★ It is the principal muscle of respiration.

1. Structure

The diaphragm has two main parts:

A. Peripheral muscular part

Muscle fibers arise from the margins of the inferior thoracic aperture.

It has 3 parts:

Part	Attachment
Sternal	Posterior surface of xiphoid process
Costal	Inner surfaces of lower 6 ribs/costal cartilages
Lumbar (vertebral)	Lumbar vertebrae through the crura

B. Central tendon

- ▼ Muscular fibers converge toward the center.
- ▼ Forms a trefoil/clover-shaped central tendon.
- ▼ The central tendon is aponeurotic.

2. Crura

The crura are muscular attachments of the lumbar part of the diaphragm.

Right crus

- ▼ Larger and longer
- ▼ Usually arises from upper 3–4 lumbar vertebrae

Left crus

▼ Smaller

▼ Usually arises from upper 2–3 lumbar vertebrae

🧠 Right = 3–4

🧠 Left = 2–3

📅 3. Major Openings of the Diaphragm

Three major openings are extremely important clinically and in exams:

Opening	Level	Main contents
Caval opening	T8	IVC
Esophageal hiatus	T10	Esophagus + vagal trunks
Aortic hiatus	T12	Aorta + thoracic duct

📍 T8 — Caval Opening

📍 Located in the central tendon.

Contains:

▼ Inferior vena cava

▼ Branches of the right phrenic nerve

▼ Lymphatic vessels

★ IVC passes through the central tendon.

📍 T10 — Esophageal Hiatus

Contains:

◆ Esophagus

◆ Anterior & posterior vagal trunks

◆ Esophageal branches of left gastric vessels

★ The esophageal hiatus is formed mainly by fibers of the right crus.

🔴 T12 — Aortic Hiatus

▼ Located posterior to the diaphragm.

Contains:

▼ Aorta

▼ Thoracic duct

▼ Azygos/hemiazygos veins

★ The aorta does not pass through the muscle of the diaphragm; it passes posteriorly through the aortic hiatus.

4. Nerve Supply

★ Phrenic Nerve — C3, C4, C5

Motor

➡ Phrenic nerve provides ALL motor innervation to the diaphragm.

🧠 "C3, 4, 5 keep the diaphragm alive!" 🫁

Sensory

Region	Nerve
♥ Central diaphragm	Phrenic nerve
● Peripheral diaphragm	Lower 6 intercostal + subcostal nerves

🩺 Clinical significance

Irritation of the phrenic nerve/central diaphragmatic pleura can produce referred pain to the shoulder because the phrenic nerve arises from C3–C5. 🤕

Function During Respiration

Inspiration

Diaphragm contracts



Domes move inferiorly



Thoracic volume increases



Intrapulmonary pressure decreases



Air enters lungs 🫁

Expiration

Diaphragm relaxes



Domes move superiorly



Thoracic volume decreases



Air leaves lungs 🫁

★ Important Anatomical Relations

The right dome is higher than the left.

◆ Right dome

Higher because of the liver.

◆ Left dome

Lower because the heart and stomach occupy part of the left side.

Surface Anatomy of the Thorax

Surface anatomy uses palpable landmarks and imaginary reference lines to locate organs, describe lesions, and perform clinical examination. 🩺

1. Important Palpable Landmarks

1 Jugular Notch / Suprasternal Notch

📍 Located at the superior border of the manubrium, between the medial ends of the clavicles.

◆ Easily palpable in the midline.

◆ Approximately at the level of T2.

🧠 Jugular notch → T2

★ 2. Sternal Angle — Angle of Louis

📍 Junction between:

Manubrium + body of sternum

➡ Forms the manubriosternal joint.

Level:

T4–T5 intervertebral disc ★

Major structures/landmarks at this level:

- ✓ 2nd costal cartilage
- ✓ Tracheal bifurcation (carina)
- ✓ Beginning and end of arch of aorta
- ✓ Bifurcation of pulmonary trunk

Clinical importance

The sternal angle is used to count ribs and intercostal spaces.

★ Find the 2nd costal cartilage at the sternal angle → then count downward.

3 Xiphoid Process

- 📌 Inferior part of sternum.
- ▼ Smallest and most variable part.
- ▼ Located in the epigastric region.
- ▼ Approximately around T9–T10, depending on position and reference.

Important relations:

- ▼ Superior border of liver
- ▼ Central tendon of diaphragm
- ▼ Inferior border of heart

⚠ Exact vertebral level can vary with respiration, age, and body position.

2. Reference Lines of the Thorax

These imaginary vertical lines help describe the location of structures during physical examination.

A. Anterior Median / Midsternal Line

- Vertical line through the midline of the sternum.

➔ Divides the anterior thoracic wall into right and left halves.

B. Midclavicular Line (MCL)

- Vertical line passing through the midpoint of the clavicle.

➔ Used frequently in surface anatomy and clinical examination.

C. Axillary Lines

There are 3 important axillary lines.

1 Anterior Axillary Line

- Extends vertically from the anterior axillary fold.

➔ The anterior fold is mainly formed by pectoralis major.

2 Midaxillary Line

- Vertical line through the middle of the axilla.

➔ Located between the anterior and posterior axillary lines.

3 Posterior Axillary Line

- Extends vertically from the posterior axillary fold.

➔ Posterior fold is formed mainly by:

Latissimus dorsi

Teres major

D. Posterior Reference Lines

1 Posterior Median Line

• Vertical line along the tips of the thoracic spinous processes.

➡ Also called the vertebral line in many clinical descriptions.

2 Scapular Line

• Vertical line passing through the inferior angle of the scapula when the upper limb is in the anatomical/neutral position.

➡ Used as a reference for posterior thoracic examination.

The transverse process of a thoracic vertebra and the tubercle of a corresponding rib meet to form the costovertebral joint, a small synovial articulation that stabilizes the rib cage and allows gliding movements during breathing.

Joint Structure and Anatomy

Transverse Costal Facet: A small smooth patch on the tip of the thoracic vertebra's transverse process.

Articular Part of Tubercle: A matching oval facet on the posterior rib tubercle that plugs into the vertebra's transverse process.

Non-Articular Part: A rough bony patch next to the facet that anchors strong stabilizing costovertebral ligaments.

What runs in the Costal Groove?

The major structures are the:

Intercostal neurovascular bundle

• Posterior intercostal vein

• Posterior intercostal artery

• Intercostal nerve

Their usual arrangement from superior → inferior is:

🧠 V-A-N

V = Vein

A = Artery

N = Nerve

🦴 Costal Grooves

The costal groove is a longitudinal groove on the inferior internal surface of a typical rib.

It is found along the:

➡ Inferior border of the internal surface of the shaft of a typical rib.

🩸 What runs in the Costal Groove?

The major structures are the:

Intercostal neurovascular bundle

🩸 Posterior intercostal vein

🩸 Posterior intercostal artery

⚡ Intercostal nerve

Their usual arrangement from superior → inferior is:

🧠 V-A-N

🧬 Relationship to Intercostal Muscles

The intercostal neurovascular bundle lies primarily between the internal intercostal and innermost intercostal muscles.

So from superficial → deep:

External intercostal muscle



Internal intercostal muscle



🔴 Vein

🔴 Artery

⚡ Nerve



Innermost intercostal muscle

Remember

The main neurovascular bundle is in the:

Neurovascular plane between the internal and innermost intercostal muscles.

Clinical Importance

Thoracentesis / Chest Tube Placement

▼ This is one of the most important clinical applications of the costal groove.

Because the main intercostal neurovascular bundle lies in the costal groove along the inferior border of the rib, inserting a needle or chest tube immediately below a rib can injure:

🔴 Intercostal artery

🔴 Intercostal vein

⚡ Intercostal nerve

✓ Therefore, procedures involving the intercostal space are generally performed just superior to the upper border of the rib below to avoid the main neurovascular bundle.

★ Example

If entering the 6th intercostal space:

✗ Avoid the inferior border of the 6th rib.

✓ Enter near the superior border of the 7th rib.

Subclavian vein = anterior to anterior scalene

Subclavian artery = posterior to anterior scalene

Brachial plexus = posterior to anterior scalene

Bronchial Venous Drainage

▼ The bronchial veins drain blood from the bronchial tree and supporting tissues of the lungs.

★ Not all blood from the lungs returns through the bronchial veins.

▼ The pulmonary circulation also receives blood from the bronchial circulation, and some of that venous blood ultimately enters the pulmonary veins.

Right vs Left Bronchial Veins

● Right bronchial vein

✓ The right bronchial vein usually drains into the Azygos vein

● Left bronchial veins

✓ The left bronchial veins usually drain into the Accessory hemiazygos vein and/or Left superior intercostal vein

◆ These ultimately communicate with the azygos system and drain toward the superior vena cava.

★ 3. The Important Exception: Pulmonary Veins

This is the part you must understand for exams.

▼ Some bronchial venous blood drains into the Pulmonary veins

✓ instead of returning directly through the systemic veins.

Why is this important?

✓ Because this creates a physiological right-to-left shunt.

✓ This blood has not passed through the pulmonary capillary bed for gas exchange.

Therefore, a small amount of deoxygenated blood mixes with oxygenated blood entering the left atrium.